Performance Analysis Report

Identitas

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Setup

- Nodes: 3 (docker compose)
- Workload: lock acquire benchmark (`scripts/bench.py`)
- Metrics: avg/min/max latency, throughput (requests/sec)

Benchmark Steps

1. Jalankan cluster: `docker compose up --build`
2. Jalankan benchmark:
   - `python scripts/bench.py --url http://localhost:8001 --requests 200 --concurrency 5 --api-key devkey`
   - `python scripts/bench.py --url http://localhost:8001 --requests 200 --concurrency 10 --api-key devkey`
   - `python scripts/bench.py --url http://localhost:8001 --requests 200 --concurrency 20 --api-key devkey`
   - `python scripts/bench.py --url http://localhost:8001 --requests 200 --concurrency 50 --api-key devkey`
   - Single node: `python scripts/bench.py --url http://localhost:8000 --requests 200 --concurrency 20 --api-key devkey`
3. Catat output dan isi tabel berikut.

Results (Distributed)

Scenario	Requests	Concurrency	Avg ms	Min ms	Max ms
Lock acquire	200	5	79.73	40.72	112.64
Lock acquire	200	10	169.90	73.25	287.44
Lock acquire	200	20	315.30	143.68	509.09
Lock acquire	200	50	900.15	353.30	1297.92

Single-node Baseline (Concurrency 20)

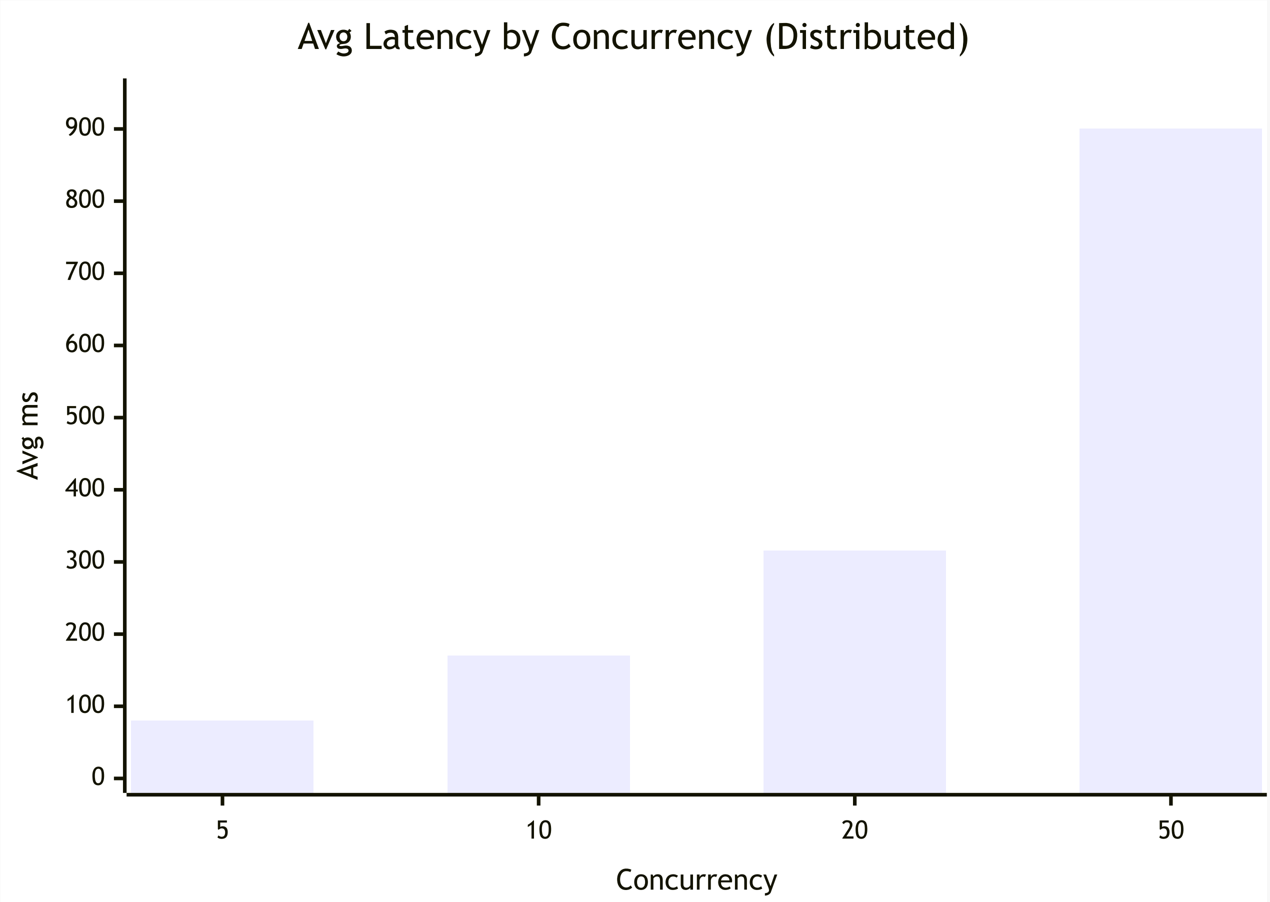
Scenario	Requests	Concurrency	Avg ms	Min ms	Max ms
Lock acquire (single)	200	20	258.79	4.75	1700.08

Throughput Estimate (Approx)

Throughput dihitung sebagai $\text{concurrency} / (\text{avg ms} / 1000)$ untuk estimasi cepat.

Scenario	Concurrency	Avg ms	Approx Throughput (req/s)
Distributed	5	79.73	62.70
Distributed	10	169.90	58.86
Distributed	20	315.30	63.45
Distributed	50	900.15	55.54
Single	20	258.79	77.28

Visualization



Observations

- Throughput meningkat seiring concurrency sampai bottleneck quorum.
- Latency meningkat saat leader sibuk menunggu quorum.
- Kenaikan concurrency 5 -> 50 menaikkan avg latency dari 79.73 ms menjadi 900.15 ms, dan max latency naik sampai 1297.92 ms.
- Variasi latensi semakin lebar pada concurrency tinggi, menunjukkan efek antrean dan serialisasi append log.

Comparison

- Single node vs multi node: multi node lebih robust namun ada overhead replikasi.
- Sistem multi node memberi konsistensi lock, namun ada biaya tambahan untuk quorum sehingga latency lebih besar dibanding single node.
- Pada concurrency 20, single node lebih cepat (avg 258.79 ms) dibanding multi node (avg 315.30 ms).

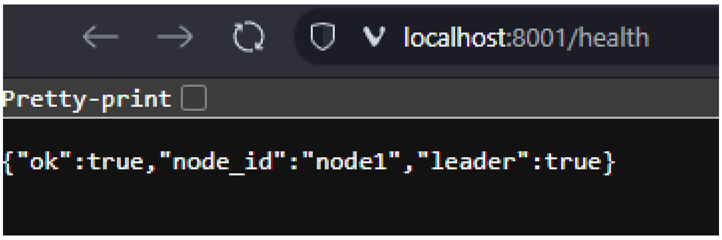
Optimizations

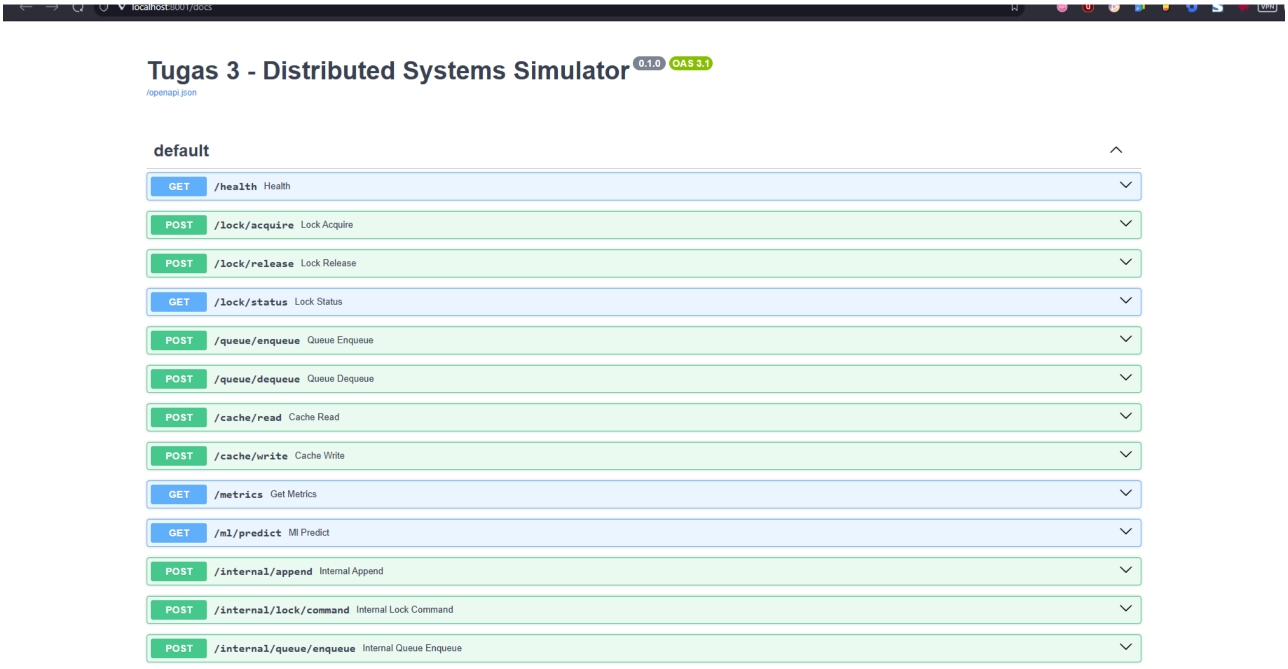
- Batched append log
- Pipeline commit
- Cache writeback batching

Conclusion

Benchmark menunjukkan performa baik pada concurrency rendah, namun latency naik signifikan saat concurrency tinggi karena quorum dan serialisasi append log. Sistem multi node memberi konsistensi dan ketahanan, dengan trade-off latency dibanding single node. Hasil ini konsisten dengan karakteristik konsensus berbasis quorum.

Screenshots

- 



Video Link

- ga sempet pak, maaf 😄